

Break Out the Silverware: Semantic Understanding of Stored Household Items

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Abstract. “Bring me a plate.” For domestic service robots, such a simple command exposes a fundamental challenge: inferring where everyday objects are stored when they are not directly visible, such as inside drawers, cabinets, or closets. While recent advances in vision and manipulation have improved robotic perception, robots still lack the commonsense reasoning needed to predict plausible storage locations. We introduce the *Stored Household Item Challenge*, a benchmark task for evaluating a robot’s ability to infer likely storage locations given a household kitchen scene and a queried item. The benchmark comprises two datasets: (1) a real-world evaluation set of 100 item–image pairs with human-annotated ground truth from participants’ kitchens, and (2) a development set of 6,500 item–image pairs with polygon-level storage annotations on public kitchen images. To address this task, we propose NOAM (Non-visible Object Allocation Model), a hybrid vision–language pipeline that converts visual input into structured natural-language descriptions of spatial context and visible containers, and then prompts a large language model to infer the most likely hidden storage location. We evaluate NOAM against random baselines, vision–language pipelines, state-of-the-art multimodal models, and human performance. NOAM substantially improves prediction accuracy and approaches human-level results, demonstrating the value of structured vision–language reasoning for cognitively capable service robots in domestic environments.

Keywords: Semantic Understanding · Commonsense Reasoning · Domestic Robotics · Large Language Models · Benchmark Datasets.

1 Introduction

Service robots are increasingly capable of navigating and manipulating objects in household environments. Yet, they remain limited in a fundamental way: commonsense reasoning about where items are typically stored when those items are not visible. For humans, such inference is effortless. When asked to fetch a mug, we do not search the refrigerator or under the sink; instead, we rely



Fig. 1. Kitchen scene shown to Gemini, with its predicted spoon storage location highlighted in the bounding box.



Fig. 2. Kitchen scene shown to GPT, with its predicted spoon storage location highlighted in the bounding box.

on deeply shared expectations about household organization. Translating this intuition into machine reasoning remains an unsolved problem.

We introduce the **Stored Household Item Challenge**, a new benchmark designed to evaluate a specific form of commonsense reasoning: predicting the likely storage location of household items that are not visible. Given an indoor kitchen scene and a queried item, the task is to infer which storage container (such as a drawer, cabinet, or closet) is most likely to contain the item, despite it being hidden from view. Unlike traditional object detection, which focuses on visible entities, this task requires reasoning about concealed items based on indirect visual cues and prior semantic knowledge. We hypothesize that household storage follows consistent patterns (for example, mugs are usually stored in upper cabinets rather than drawers) that can be learned and generalized.

Current benchmarks overlook this reasoning dimension. Vision-Language Models (VLMs) and object detectors perform well on visible items but fail when asked to infer the invisible. For instance, we asked Gemini and GPT to identify where a spoon is most likely stored, using the following prompt:

In the given image, which drawer or cabinet is the most likely place to store a spoon? Return the image with the exact drawer or cabinet clearly marked. Do not modify any other part of the image.

As shown in Figures 1 and 2, both models struggled to select a plausible container. To our knowledge, no prior structured benchmark exists for this ability. Existing datasets lack labels for expected-but-not-visible items, and current models are not trained to reason about functional storage norms or household organization. This gap highlights both a scientific challenge and a practical barrier for embodied agents operating in real homes.

To support this task, we construct two datasets: (1) a 6,500-pair development set based on real kitchen scenes (SUN dataset), with container annotations by humans; and (2) a 100-example evaluation set collected from real homes. These datasets enable scalable training and robust testing across varied environments.

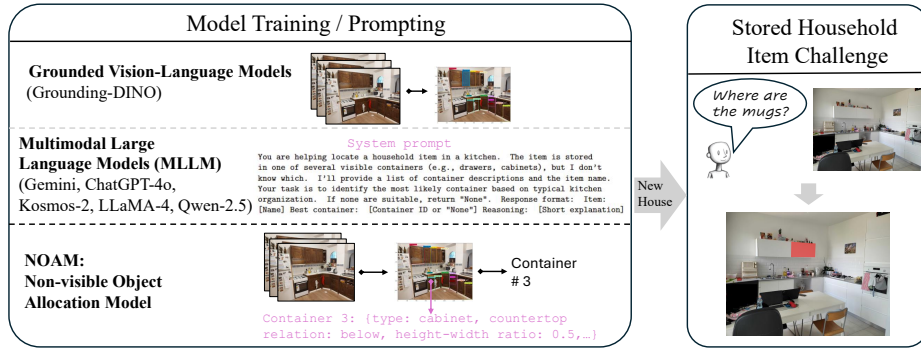


Fig. 3. Depiction of the Stored Household Item Challenge (right). We compare the performance of various models on this task (left), including grounded vision-language models, multimodal LLMs, and our model, NOAM.

We benchmark random baselines, VLMs, Multimodal Large Language Models (MLLMs), and human predictions (Figure 3, top and middle left).

We also introduce **NOAM** (Non-visible Object Allocation Model), a novel language-based pipeline that reframes the task as structured text-based reasoning (Figure 3, bottom left). By describing scenes and candidate storage options in natural language and querying LLMs, NOAM significantly outperforms VLMs, though still below human performance. This suggests that language-driven inference offers a promising path for equipping agents with commonsense reasoning skills. To support reproducibility, we make the code, evaluation scripts, and datasets publicly available at <https://github.com/michaelalr/thesis>.

Rather than chasing marginal detection gains, our aim is to formalize a reasoning capability absent in today’s models: context-aware inference about what is unseen but likely. The *Stored Household Item Challenge* captures this missing skill that is crucial for robots in cluttered, unfamiliar spaces. As LLMs evolve, we expect progress, but the challenge will remain a meaningful test of real-world understanding.

To summarize, the contributions of this work are threefold:

1. New challenge: the Stored Household Item Challenge.
2. An annotation pipeline and dataset of item-image pairs grounded in real household scenes.
3. NOAM (Non-visible Object Allocation Model), a language-based model that reframes visual inference as a structured textual reasoning task. We compare it against baselines, including random choice, VLM, MLLMs and humans.

2 Related Work

Our research builds on domestic service robotics and visual scene understanding, addressing the specific challenge of reasoning about hidden items - a capability largely absent in current systems.

Domestic Service Robots Service robots have evolved from vacuuming to complex tidying and organization tasks [7, 11, 22]. Prior approaches often learn user preferences via GNNs or collaborative filtering [1, 10, 25], or utilize knowledge-based ontologies [15]. However, these methods typically focus on manipulating objects on *visible* surfaces like tables [8, 20, 23, 28]. While some works address object search, they often assume partial observability [9] or require prior maps of container contents [14]. In contrast, our approach infers the location of completely hidden items in closed containers without requiring prior environmental maps, aiming to generalize storage norms for immediate deployment in unfamiliar homes.

Visual Perception and Reasoning Effective robot operation requires holistic scene understanding, spanning scene classification [18, 24] and SLAM for geometric mapping [3, 6]. While recent object detection advances leverage deep learning [29] and open-set text prompts (e.g., Grounding-DINO [16]), they remain limited to visible entities. Techniques like Concealed Object Detection (COD) [5] or occlusion handling [21] address camouflaged or partially visible objects, but do not infer locations of items fully enclosed in storage. We leverage state-of-the-art segmentation tools like SAM [12] and Grounding-DINO within a VLM-driven pipeline (NOAM) to bridge this gap, moving from detecting the visible to reasoning about the invisible.

3 The Stored Household Item Challenge

We introduce the **Stored Household Item Challenge**, a benchmark for evaluating *Commonsense Storage Prediction* in domestic environments. This task requires reasoning about *non-visible* items. A system is to infer plausible hidden locations using visual context, semantic priors, and typical storage conventions. Although the challenge is defined for domestic environments in general, we instantiate it in this study using kitchen scenes, which provide a structured and common setting for evaluating storage-related commonsense. For example, given the query “mug”, the model should prefer an upper cabinet over a lower drawer, even if the mug is not visible.

Input: A single RGB image of a kitchen scene and the name of a common household item (e.g., “mug”, “cutting board”).

Output: The most likely storage container *instance* (e.g., a specific drawer or cabinet) in the image where the item is stored. The prediction must be grounded in a 2D polygon corresponding to a visible container.

This task involves three commonsense reasoning challenges:

- **Reasoning Beyond Visibility:** Most vision systems detect only visible objects. This task requires inferring hidden item locations using indirect cues and commonsense knowledge.
- **Semantic Understanding and Commonsense Priors:** Success depends on functional knowledge of household norms combined with visual context.
- **Grounded, Actionable Output:** Predictions must be specific and physically actionable. Rather than generic guesses like “a cabinet,” the model must identify a concrete container instance suitable for robotic interaction.

To date, no existing dataset targets reasoning about hidden storage in realistic household scenes, and VLMs often default to literal descriptions of what is visible. By offering a structured task, curated data, and measurable evaluation criteria, this challenge fills a critical gap and provides a testbed for evaluating commonsense visual reasoning in service robotics.

4 Datasets and Data Collection

To predict where household items are typically stored, we required a dataset capturing commonsense storage behavior. Specifically, where items are likely kept out of sight in containers. As no existing dataset supported this task, creating a new one posed two challenges. First, in terms of **privacy**, domestic images often contain personal content that is difficult to anonymize and share ethically. Second, it is hard to **scale**. Gathering a large and diverse set of labeled images from volunteers is logistically challenging, making small-scale collection feasible but broad-scale coverage impractical. To address these challenges, we adopted a dual-dataset strategy:

- A **crowdsourced development dataset** using public kitchen images, supporting design, tuning, and analysis.

- A smaller **real-world evaluation dataset** collected from participants’ homes, used for final testing.

The development set was built from the SUN dataset [27, 26]. Annotators recruited via the Upwork freelance platform³ selected the most likely storage container for a given item in each kitchen image. Figure 4 shows the annotation interface. We cleaned the data, removed duplicates, assessed annotator agreement, and curated 6,500 high-quality item-image pairs with polygon-level annotations.

The evaluation set comprises 100 item–image pairs collected from real kitchens, each annotated with the actual storage location of the queried item, serving as the ground-truth reference for model assessment.

To constrain variability, we focused on kitchens and selected 15 household items spanning common and uncommon examples: *bottle opener*, *Tupperware containers*, *dish towels*, *cutting board*, *bowl*, *spices*, *spoon*, *mug*, *plate*, *pot*, *pan*, *cutting knife*, *cooking oil*, *screwdrivers*, and *painkillers*.

An IRB from *redacted for anonymity* approved data collection involving human participants, covering both datasets. Additional details of the annotation process are provided in the supplementary material.

4.1 Dataset Preprocessing

We applied Grounding-DINO [16] and SAM [13] to segment all visible storage containers. Each development image contained ~ 16 containers on average; each evaluation image, ~ 19 . These segmentations provide visual cues for detection and support automated answer verification, avoiding subjective free-form judgments.

4.2 Development Dataset: Crowdsourced Annotations

From 1,746 kitchen images in SUN, we selected 1,656 with at least three detectable containers. From the 15 household items, two were held out for generalization tests. For each of the remaining 13 items, we sampled 500 images from the SUN kitchen scenes yielding a total of 6,500 item–image pairs.

Three annotators (two from the U.S., one from Ireland) used a custom web tool to label images. Each image showed an item name and marked containers, as shown in Figure 4. Responses were stored in a database. Each annotator contributed 2,860 annotations (1,040 overlapping). Annotations were done in batches and manually reviewed. We collected 8,580 total annotations, from which the 6,500 development pairs were drawn. To assess quality, 16% of examples per item were triple-labeled. Agreement was computed using Fleiss’ Kappa, detailed in Section 4.4.

³ <https://www.upwork.com/>

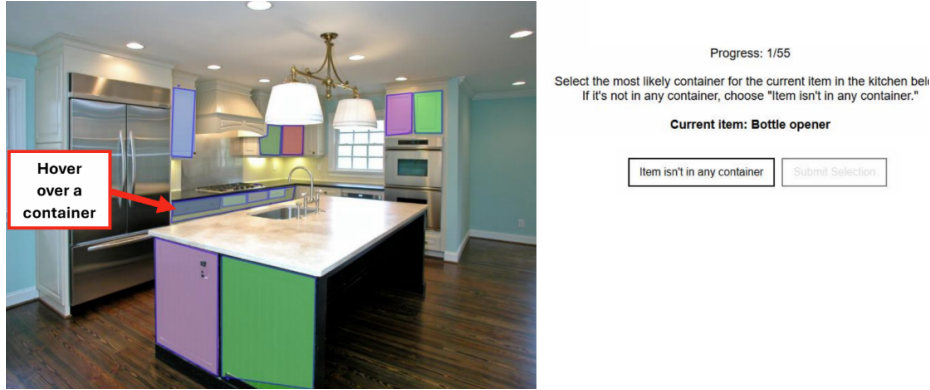


Fig. 4. A screenshot from the annotation tool used to collect human-labeled data efficiently.

4.3 Evaluation Data: Real-World Objects

We collected 100 item-image examples from 74 participant kitchens. Participants labeled storage locations for *bottle openers*, *Tupperware containers*, *painkillers*, and *screwdrivers*. The last two were excluded from the development set to test generalization. In addition, some participants labeled an extra item of their choice, beyond the four predefined categories. Participants were instructed to exclude sensitive content, and all images were manually reviewed for identifying details. This dataset was used both to validate annotator reliability and as ground truth for model evaluation.

4.4 Data Cleaning and Processing

We applied the following procedures to ensure data quality:

1. Removed duplicates and conflicting responses.
2. Consolidated multiple annotations:
 - Majority vote was used when two annotators agreed.
 - Otherwise, one label was selected at random.
3. Split data into development and evaluation sets.

Overall inter-annotator agreement was moderate, with Fleiss’ Kappa of $\kappa = 0.354$ on the development set and $\kappa = 0.372$ on the evaluation set. Agreement varied across items: objects with standardized storage locations (e.g., bottle openers, $\kappa = 0.494$; plates, $\kappa = 0.478$) exhibited higher consistency, whereas more subjective categories such as Tupperware showed lower agreement ($\kappa = 0.272$). These results reflect both the inherent ambiguity of the task and meaningful human regularities, indicating that annotators rely on shared commonsense rather than random choice when inferring hidden household item locations.

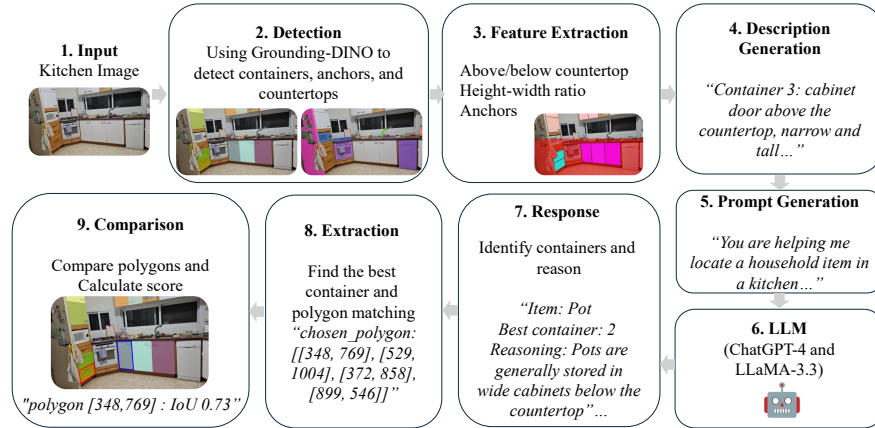


Fig. 5. An overview of our NOAM pipeline: from an input image (1) through container detection (2), feature extraction (3), textual explanation (4), prompt generation (5), through LLM processing (6) to response (7) and value extraction (8) and overall evaluation (9).

5 NOAM: Non-visible Object Allocation Model

Our goal is to enable VLMs, and later on service robots, to reason about storage locations with accuracy approaching or exceeding human-level performance as measured on our dataset ($\sim 30\%$). To this end, we reformulate the task from an image-text multimodal problem into a purely language-based one, casting it as a natural language understanding challenge.

Our approach, the Non-visible Object Allocation Model (NOAM), is outlined in Figure 5. For each container detected in the scene, we extract a set of visual and spatial features and translate them into clear, natural language descriptions (steps 2–3). These descriptions are incorporated into carefully structured prompts that instruct a language model (e.g., ChatGPT) to infer the most probable storage location for a given item (steps 4–5). The model then reasons to select the most probable container(s) (step 7). We extract the predicted container ID, defaulting to the first option when multiple candidates are returned, and map it to the corresponding polygon in the image (step 8). Finally, spatial accuracy is evaluated using Intersection over Union (IoU) with the ground truth (step 9).

Feature Extraction To represent each container in an image, we extracted features relevant to reasoning, including: container type (e.g., drawer, cabinet door) and confidence score, position relative to the countertop (above/below), neighboring containers and anchor objects (e.g., sink, oven), closest container to each anchor, aspect ratio and shape, and other contextual cues.

Some features were computed directly from container polygons, while others required re-applying Grounding-DINO and SAM to detect countertops and

anchor objects. Each container was then represented as a row in a *container information table*, initially containing image path and polygon data. We augmented the table with the extracted features through the following steps:

1. Assign global and local container IDs.
2. Extract and assign labels and confidence scores.
3. Resolve ambiguous labels (e.g., “drawer cabinet door”) using context from neighboring containers.
4. Compute container dimensions and aspect ratio.
5. Determine spatial relation to the countertop.
6. Identify neighboring containers and anchor objects.
7. Measure distances and angles from each container to all anchors to determine the closest container to each anchor.

Natural Language Description This stage involves two main steps: first, selecting the most informative and relevant features to include in each container’s description, with a focus on attributes that help the language model infer spatial and categorical relationships; and second, converting these features into clear and intuitive natural language. Rather than using raw numerical values (e.g., aspect ratios), we express features qualitatively. For example, using terms like “wider than tall”, “square-like”, and “close to the sink” empirically improved model performance. This strategy aligns with prior findings that language-based tasks are generally more tractable for large language models than compositional or arithmetic reasoning tasks [4, 19].

Prompt Design and Task Framing To guide the language model in selecting the most likely storage container for a given item, we explored multiple prompt strategies using ChatGPT-4 [17] and LLaMA-3.3 [2]. Each prompt presented a list of container descriptions and required the model to return the most probable container ID, or “None” if none were suitable.

We experimented with three prompt types: (1) Instructional prompts, framing the model as a service robot reasoning about each item; (2) Story-based prompts, encouraging narrative completion grounded in the container descriptions; and (3) Structured system-user prompts, separating global task instructions (system prompt) from instance-specific inputs (user prompt), which yielded the best results. The complete prompt design is provided in the supplementary material.

6 Experimental Setup

We evaluated our method on the real-world dataset (Section 4.3), comparing two model categories: vision–language models (VLMs) and multimodal large language models (MLLMs). Baselines include Grounding-DINO, Kosmos-2, Gemini, GPT-4o, LLaMA-4, and Qwen-2.5, all benchmarked against NOAM.

Model configuration was tuned on the development set (Section 4.2), using 5% (370 of 6,500 item-image pairs). Each model was prompted with a household item and asked to predict its most likely storage location (e.g., drawer or cabinet door) by returning a bounding box. Performance was measured using Intersection over Union (IoU) and accuracy at a fixed threshold.

6.1 Evaluation Metrics

Models were evaluated on both development and test sets using:

- **Accuracy:** the proportion of predictions with $\text{IoU} \geq 0.5$.
- **IoU:** the overlap between predicted and ground-truth bounding boxes.

The IoU threshold of 0.5 was selected based on human-labeled quality analysis, where correct or partially correct predictions averaged 0.54 IoU. This value balances strictness with human interpretability. For Grounding-DINO, we additionally report results at a stricter $\text{IoU} \geq 0.95$ (still allowing for some misalignment). Full details on how these thresholds were derived are available in the supplementary material.

6.2 Baselines

We compare NOAM against the following methods:

Random. A container was randomly selected from the set of detected drawers and cabinet doors for each image as provided by Grounding-DINO during the annotation phase.

Grounding-DINO and SAM. A vision-based pipeline detecting candidate containers and segmenting them, with the top-confidence prediction selected.

Grounding-DINO (No Item Prompt). The same pipeline without conditioning on the queried item, isolating the effect of item-specific prompts.

MLLMs. We evaluate Kosmos-2, Gemini, GPT-4o, LLaMA-4, and Qwen-2.5, each prompted to predict the most likely storage location given an image and an item.

Full prompts, implementation details, hyperparameters, and parsing procedures for all baselines as well as computing environment details are provided in the supplementary material.

7 Results and Analysis

Real-World Evaluation Dataset We begin by reporting the overall performance of the various algorithms on the real-world evaluation dataset. As shown in Table 1, NOAM achieves the highest accuracy and IoU among all of the evaluated models, outperforming random, vision-based, and multimodal baselines.

For follow-up statistical tests, we conducted a one-way ANOVA on accuracy scores, followed by post-hoc pairwise comparisons using the Bonferroni correction. The results confirmed a significant effect of model type on performance ($p <$

Table 1. Evaluation Set Accuracy and Average IoU for Various Models.

Model	Accuracy (%)	Average IoU
Human Annotator 1 (IoU = 1)	38.00	0.380
Human Annotator 2 (IoU = 1)	27.00	0.271
Human Annotator 3 (IoU = 1)	36.00	0.361
Random (IoU = 1)	6.00	0.062
Grounding-DINO (IoU = 1)	13.00	0.188
Grounding-DINO (IoU ≥ 0.95)	17	0.188
Grounding-DINO - no item in prompt (IoU = 1)	10	0.117
Kosmos-2 (IoU ≥ 0.5)	4.00	0.042
Gemini-1.5-flash (IoU ≥ 0.5)	3.00	0.034
Gemini-2.5-flash (IoU ≥ 0.5)	1.00	0.027
GPT-4o API (IoU ≥ 0.5)	8.00	0.082
LLaMA-4 (IoU ≥ 0.5)	1.00	0.094
Qwen-2.5 (IoU ≥ 0.5)	5.00	0.091
NOAM LLaMA-3.3 (IoU = 1)	23.00	0.232
NOAM GPT-4 (IoU = 1)	23.00	0.23

0.05). Specifically, NOAM GPT-4 significantly outperformed Gemini-1.5-flash, Gemini-2.5-flash, and Kosmos-2 after Bonferroni correction. Differences with other models, including GPT-4o, Grounding-DINO, LLaMA-3.3, and LLaMA-4, were not statistically significant. NOAM also did not differ significantly from human annotators.

While NOAM did not reach human-level performance, its accuracy (23%) is notably closer to that of the least accurate human annotator (27%) than the gap between that annotator and the next-best human (36%). Further, the pairwise comparison between NOAM and the least accurate human annotator did not reach statistical significance ($p = 0.006, 0.46, 0.026$ for annotators 1, 2, and 3 respectively, before the Bonferroni correction and $p = 0.67, 1.0, 1.0$ after the correction). This result is encouraging and suggests that the pipeline of NOAM has strong potential to soon reach human-level performance, especially as new VLMs and MLLMs become available for integration.

Development Dataset Table 2 presents results on the annotated development set. While our focus is on evaluation performance, development scores help reveal whether models learned meaningful patterns. For instance, Kosmos-2 underperforms even on the development set (below random) suggesting limited capacity or misalignment with the task rather than overfitting. In contrast, NOAM performs strongly, reflecting effective task-specific learning. Human scores are based on overlapping annotations used for agreement analysis and do not reflect full dataset performance.

Table 2. Development Set Accuracy and Average IoU for Various Models.

Model	Accuracy (%)	Average IoU
Human Annotator (Estimated)	35.83	0.361
Random (IoU = 1)	7.55	0.083
Grounding-DINO (IoU = 1)	5.89	0.112
Grounding-DINO (IoU ≥ 0.95)	9.49	0.112
Grounding-DINO - no item in prompt (IoU = 1)	11.35	0.128
Kosmos-2 (IoU ≥ 0.5)	0.46	0.013
Gemini-1.5-flash (IoU ≥ 0.5)	8.11	0.083
Gemini-2.5-flash (IoU ≥ 0.5)	2.43	0.03
GPT-4o API (IoU ≥ 0.5)	13.78	0.138
LLaMA-4 (IoU ≥ 0.5)	3.24	0.071
Qwen-2.5 (IoU ≥ 0.5)	7.57	0.099
NOAM LLaMA-3.3 (IoU = 1)	23.51	0.244
NOAM GPT-4 (IoU = 1)	28.11	0.287

8 Discussion

NOAM demonstrates strong reasoning abilities, but several aspects warrant further development:

Efficiency and Scalability NOAM’s average inference time is approximately 13 seconds per image. The main bottlenecks are Grounding-DINO, which accounts for about 10 seconds, and the LLM API call (e.g., ChatGPT), which takes around 2.9 seconds. Using the LLaMA-3.3 (via Together-AI API) slightly increases total inference time to roughly 15.9 seconds per image.

While this runtime is acceptable for research and offline applications, real-world deployment would require further optimization. Initial experiments replacing Grounding-DINO with a lightweight detector reduced detection time to under 1 second. Future directions for improving efficiency include model distillation, prompt caching, and the use of smaller, fine-tuned modules to support near real-time inference.

Generalization and Dataset Diversity Performance is expected to vary by image domain. Expanding benchmarks to other room types (e.g., bedrooms, industrial scenes) and adapting anchors and prompts will test robustness.

Embodied and Interactive Reasoning NOAM currently uses static images. Extending it to multi-view or interactive setups could reduce occlusion errors and support belief updates during exploration.

Model Robustness LLM outputs vary by prompt and model. We plan to fine-tune VLM or MLLM on our dataset to improve consistency and compare against zero-shot performance.

Richer Priors and Personalization Integrating spatial priors (e.g., furniture hierarchies) and user-specific data (e.g., gestures or preferences) may enhance practical accuracy and personalization.

Practical Metrics IoU alone is insufficient for real-world robotics. Future work will incorporate task-based metrics (e.g., success rate, energy use), and propose container selection heuristics to improve robustness to detection errors.

9 Conclusion

This paper introduces the Stored Household Item Challenge, a novel problem designed to evaluate semantic spatial reasoning about the likely locations of non-visible objects in household environments. Unlike conventional object detection tasks, this challenge targets a critical but underexplored capability: inferring hidden item locations based on context and commonsense knowledge. We present two new datasets of labeled item-image pairs featuring concealed storage scenarios, along with a suite of baseline evaluations. To tackle this task, we further propose NOAM (Non-visible Object Allocation Model), a vision-to-language pipeline that reformulates storage prediction as a structured natural language inference problem. Experimental results demonstrate that NOAM significantly outperforms vision-only baselines, though still not comparable to human-level reasoning capabilities.

This work opens several directions for future research. For example, NOAM currently relies on static descriptions and a single scene view. Future models could benefit from richer inputs, such as sequential observations, user preferences, or multimodal feedback. Expanding the spatial reasoning framework to include hierarchical priors or soft constraints (e.g., “items are often above counters”) could further improve performance. Additionally, integrating this task into interactive, embodied agents may allow systems to refine predictions through exploration and reinforcements, enhancing robustness in real-world settings.

Although this paper focuses on kitchens, the underlying reasoning principles generalize to a wide range of domestic and industrial environments. We plan to expand the dataset and task to include additional spaces and a broader set of item categories. By advancing models of hidden object reasoning, this work contributes to building service robots that can function effectively in unfamiliar, dynamic, and complex human environments.

References

1. Abdo, N., Stachniss, C., Spinello, L., Burgard, W.: Robot, organize my shelves! Tidying up objects by predicting user preferences. In: 2015 IEEE International Conference on Robotics and Automation (ICRA). pp. 1557–1564. IEEE, Seattle, WA, USA (May 2015)
2. et al., A.G.: The llama 3 herd of models. arXiv: 2407.21783 (2024), <https://arxiv.org/abs/2407.21783>
3. Cadena, C., Carlone, L., Carrillo, H., Latif, Y., Scaramuzza, D., Neira, J., Reid, I., Leonard, J.J.: Past, Present, and Future of Simultaneous Localization and Mapping: Toward the Robust-Perception Age. *IEEE Transactions on Robotics* **32**(6), 1309–1332 (Dec 2016)
4. Dziri, N., Lu, X., Sclar, M., Li, X.L., Jiang, L., Lin, B.Y., Welleck, S., West, P., Bhagavatula, C., Le Bras, R., et al.: Faith and fate: Limits of transformers on compositionality. *Advances in Neural Information Processing Systems* **36**, 70293–70332 (2023)
5. Fan, D.P., Ji, G.P., Cheng, M.M., Shao, L.: Concealed Object Detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(10), 6024–6042 (Oct 2022)
6. Garg, S., Sünderhauf, N., Dayoub, F., Morrison, D., Cosgun, A., Carneiro, G., Wu, Q., Chin, T.J., Reid, I., Gould, S., Corke, P., Milford, M.: Semantics for robotic mapping, perception and interaction: A survey. *Foundations and Trends® in Robotics* **8**(1–2), 1–224 (2020)
7. Hasegawa, S., Taniguchi, A., Hagiwara, Y., Hafi, L.E., Taniguchi, T.: Integrating probabilistic logic and multimodal spatial concepts for efficient robotic object search in home environments. *SICE Journal of Control, Measurement, and System Integration* **16**(1), 400–422 (2023)
8. Hu, Y., Zhang, Z., Liu, R., Wyder, P., Lipson, H.: Knolling bot: A transformer-based approach to organizing a messy table. arXiv: 2310.04566 (2023)
9. Kant, Y., Ramachandran, A., Yenamandra, S., Gilitschenski, I., Batra, D., Szot, A., Agrawal, H.: Housekeep: Tidying Virtual Households Using Commonsense Reasoning. In: *Computer Vision – ECCV 2022*, pp. 355–373. Springer Nature Switzerland, Cham (2022)
10. Kapelyukh, I., Johns, E.: My house, my rules: Learning tidying preferences with graph neural networks. In: 5th Conference on Robot Learning. *Proceedings of Machine Learning Research*, vol. 164, pp. 740–749. PMLR (08–11 Nov 2022)
11. Kim, J., Mishra, A.K., Limosani, R., Scafuro, M., Cauli, N., Santos-Victor, J., Mazzolai, B., Cavallo, F.: Control strategies for cleaning robots in domestic applications: A comprehensive review. *International Journal of Advanced Robotic Systems* **16**(4) (Jul 2019)
12. Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollár, P., Girshick, R.: Segment anything. arXiv:2304.02643 (2023)
13. Kirillov, A., Mintun, E., Ravi, N., Mao, H., Rolland, C., Gustafson, L., Xiao, T., Whitehead, S., Berg, A.C., Lo, W.Y., Dollár, P., Girshick, R.: Segment anything. In: 2023 IEEE/CVF International Conference on Computer Vision (ICCV). pp. 3992–4003 (2023). <https://doi.org/10.1109/ICCV51070.2023.00371>
14. Kurenkov, A.: Manipulation and Reasoning Methods for Embodied Object Search. Ph.D. thesis, Stanford University (2023)

15. Liu, S., Tian, G., Zhang, Y., Zhang, M., Liu, S.: Service planning oriented efficient object search: A knowledge-based framework for home service robot. *Expert Systems with Applications* **187** (Jan 2022)
16. Liu, S., Zeng, Z., Ren, T., Li, F., Zhang, H., Yang, J., Jiang, Q., Li, C., Yang, J., Su, H., Zhu, J., Zhang, L.: Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In: *Computer Vision – ECCV 2024: 18th European Conference, Milan, Italy, September 29–October 4, 2024, Proceedings, Part XLVII*. pp. 38–55. Springer-Verlag, Berlin, Heidelberg (2024). https://doi.org/10.1007/978-3-031-72970-6_3, https://doi.org/10.1007/978-3-031-72970-6_3
17. OpenAI: Gpt-4 technical report. *ArXiv abs/2303.08774* (2023), <https://arxiv.org/abs/2303.08774>
18. Patel, T.A., Dabhi, V.K., Prajapati, H.B.: Survey on Scene Classification techniques. In: *2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS)*. pp. 452–458. IEEE, Coimbatore, India (Mar 2020)
19. Qian, J., Wang, H., Li, Z., Li, S., Yan, X.: Limitations of language models in arithmetic and symbolic induction. In: *The 61st Annual Meeting Of The Association For Computational Linguistics* (2023)
20. Ramrakhya, R., Kembhavi, A., Batra, D., Kira, Z., Zeng, K.H., Weihs, L.: Seeing the unseen: Visual common sense for semantic placement. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 16273–16283 (2024)
21. Saleh, K., Szenasi, S., Vamossy, Z.: Occlusion Handling in Generic Object Detection: A Review. In: *2021 IEEE 19th World Symposium on Applied Machine Intelligence and Informatics (SAMI)*. pp. 000477–000484. IEEE, Herl’any, Slovakia (Jan 2021)
22. Shah, R., Jiang, Y., Hart, J., Stone, P.: Deep R-Learning for Continual Area Sweeping. In: *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. pp. 5542–5547. IEEE (Oct 2020)
23. Shiba, T., Ono, T., Tamukoh, H.: Object Search and Empty Space Detection System for Home Service Robot. *Proceedings of International Conference on Artificial Life and Robotics* **28**, 400–403 (Feb 2023)
24. Tong, Z., Shi, D., Yan, B., Wei, J.: A Review of Indoor-Outdoor Scene Classification. In: *2nd International Conference on Control, Automation and Artificial Intelligence (CAAI 2017)*. Sanya, China (2017)
25. Wu, J., Antonova, R., Kan, A., Lepert, M., Zeng, A., Song, S., Bohg, J., Rusinkiewicz, S., Funkhouser, T.: Tidybot: personalized robot assistance with large language models. *Autonomous Robots* **47**(8), 1087–1102 (Nov 2023)
26. Xiao, J., Ehinger, K.A., Hays, J., Torralba, A., Oliva, A.: Sun database: Exploring a large collection of scene categories. *Int. J. Comput. Vision* **119**(1), 3–22 (Aug 2016)
27. Xiao, J., Hays, J., Ehinger, K.A., Oliva, A., Torralba, A.: Sun database: Large-scale scene recognition from abbey to zoo. In: *IEEE Computer Society Conference on Computer Vision and Pattern Recognition* (2010)
28. Xu, Y., Hsu, D.: “tidy up the table”: Grounding common-sense objective for tabletop object rearrangement. *arXiv preprint arXiv:2307.11319* (2023)
29. Zhao, Z.Q., Zheng, P., Xu, S.T., Wu, X.: Object Detection With Deep Learning: A Review. *Neural Networks and Learning Systems* **30**(11), 3212–3232 (Nov 2019)